

Wang Yao Architectural Portfolio

2017-2021

Q CITY International Young Designers Competition

碳中和之塔

Tower of carbon neutralization

在那个机器轰鸣、
蒸汽纷腾的工业年代的激情消却时，
留下的会是什么？
——沧桑的砖墙与混凝土早已失去了往昔的骄傲，
变得粗糙不堪。
唐河边遥望夕阳的冷却塔便是其中的一员，
曾经它的胸腔中奔涌着沸腾与冰冷的呐喊，
如今却早已是人去楼空，
只有它厚实的外壁还在诉说着一个时代的记忆。

Roar in that machine,
When the passion of the steamy industrial age fades,
What will be left?
——The vicissitudes of brick walls and concrete have long lost the pride of the past,
Become rough and unbearable.
One of them is the cooling tower looking at the sunset by the Tang River.
There used to be boiling and cold shouts in its chest,
Nowadays, people are already emptying the building,
Only its thick outer wall still tells the memory of an era.

BACKGROUND

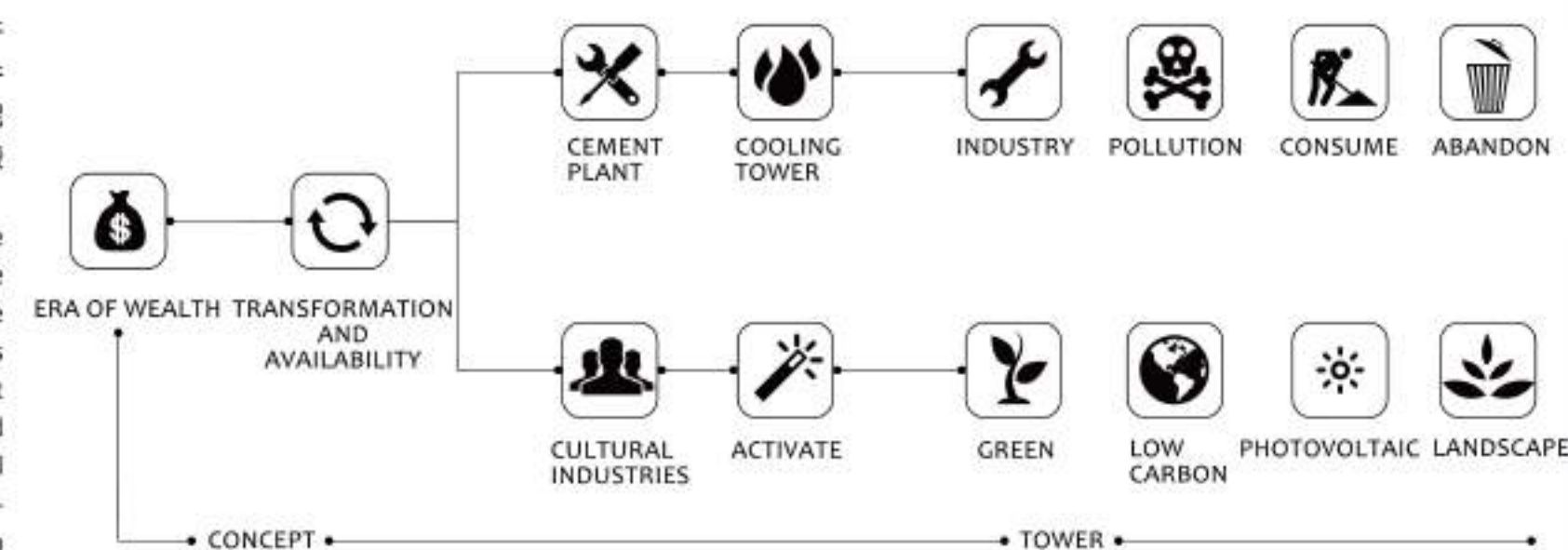


DESIGN DESCRIPTION

位于唐山市陡河边的冷却塔是启新水泥厂生产线的一部分，启新水泥厂始建于1889年，是我国最早水泥企业。生产的水泥占国内市场份额一度超过了90%并远销海外，为我国水泥工业和社会建设作出了巨大贡献。到2008年，启新水泥厂已完成了其特有的历史使命，为了响应国家产业升级、节能降耗号召而停产搬迁，冷却塔也一度闲置至今。

The cooling tower located by the Douhe River in Tangshan City is part of the production line of Qixin Cement Plant, which was founded in 1889 and is the earliest cement company in my country. The cement produced once accounted for more than 90% of the domestic market share and was exported overseas, making great contributions to my country's cement industry and social construction. By 2008, Qixin Cement Plant had completed its unique historical mission. In response to the national call for industrial upgrading, energy conservation and consumption reduction, it had suspended production and moved, and the cooling tower has been idle for a while.

ANALYSIS

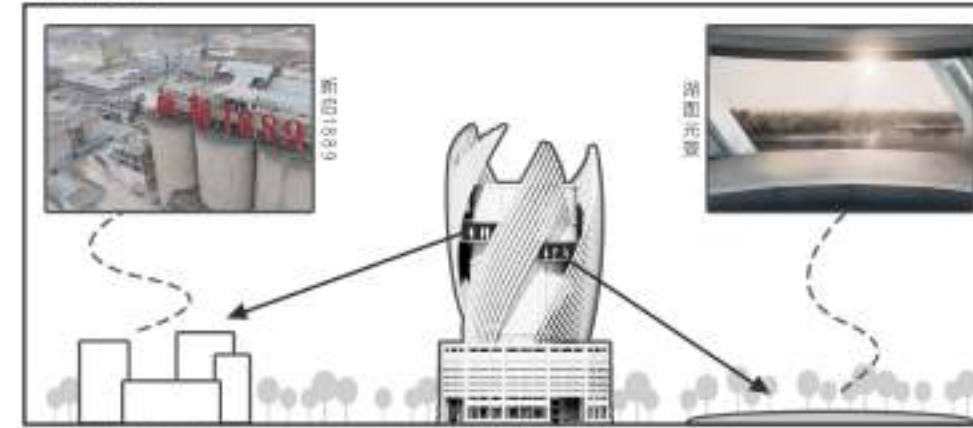


BASE INTRODUCTION

The site is located on the bank of the Tang River, next to the 1889 Cultural and Creative Industrial Park, with railroad tracks passing through on one side, and the train full of memories of the times drove far away, bidding farewell to the hustle and bustle of the industrial age, and bringing a new era of green technology and technology.



VISION



ENERGY SAVING ANALYSIS



2021年4月，习近平主席在联大会上向世界作出庄重承诺，中国将力争2030年前实现碳达峰、2060年前实现碳中和。从碳达峰到碳中和，中国设定的过渡期仅为30年，减碳之路充满压力和挑战。我国建筑面积规模位居世界第一，每年仅建筑“运营碳排放”就达到21亿吨，约占我国碳排放总量的20%，是名副其实的碳排放“大户”。因此，建筑行业的节能减碳是我国实现碳达峰、碳中和目标的关键一环。为了降低建筑碳排放，本次建筑设计我们引入“光电建筑”（PV BUILDING）概念，探索新型光伏建材与传统建筑的结合之道，将建筑节能愿景由超低能耗建筑向正能源建筑推进。

In April 2021, President Xi Jinping made a solemn promise to the world at the UN General Assembly that China will strive to achieve carbon peaks by 2030 and carbon neutrality by 2060. From carbon peak to carbon neutrality, the transition period set by China is only 30 years, and the road to carbon reduction is full of pressure and challenges. my country ranks first in the world in terms of floor space. The annual "operational carbon emissions" from buildings alone reach 2.1 billion tons, accounting for about 20% of my country's total carbon emissions. It is a veritable "larger" of carbon emissions. Therefore, energy-saving and carbon-reduction in the construction industry is a key part of my country's goal of achieving carbon peak and carbon neutrality. In order to reduce building carbon emissions, we introduced the concept of "PV BUILDING" in this building design to explore the combination of new photovoltaic building materials and traditional buildings, and promote the vision of building energy efficiency from ultra-low energy buildings to positive energy buildings.



碳中和之塔 Tower of carbon neutralization



Tower of carbon neutralization

We captured its sorrow and loneliness, and injected it with new-age technology and modeling: replacing part of the siding with a light glass curtain wall, reflecting the sparkling waves of the Tang River; and setting two locations on the outer side of the tube wall. The viewing platform, one facing the Tang River, is for people to admire the beauty of the river, and the other is facing the old cement factory site, which reminds people of the glory of the industrial era. Visitors can slowly ascend the circular staircase inside, watching the changing times, watching the thick stone slabs next to them, to the clear and translucent curtain wall, watching the transformation from the industrial age to modernization, what have we gained? As a result, we wound a circle of light "yarn" around the cooling tower, all composed of photovoltaic panels, like fresh spring bamboo shoots emerging from the ground, and like a pair of strong wings rising in the wind, which produces Electricity can supply the daily life of citizens, so that cooling towers have also been given a new mission in this era.



Diagram illustrating the wind circulation system. It shows a cross-section of a building with a central atrium and multiple levels. Arrows indicate the flow of air from the top levels down to the ground floor, creating a natural ventilation path. The diagram also shows the layout of the building's floors and the central atrium.

■ **环绕流线**
 随着楼层向上, 空气逐渐上升, 同时建筑内部凉风环境上升, 行人随着不断上升的螺旋楼梯上升, 逐渐享受凉风系统。

■ **Circumfluence streamline**
 Along with the winding spiral staircase, around the inner surface of the building pedestrians gradually rise with the rising spiral staircase, and gradually enjoy more style.

- Circumfluence streamline

Along with the winding spiral staircase, around the inner surface of the building pedestrians gradually rise with the rising spiral staircase, and gradually enjoy more style.

■ 漫步流线

■ WALKING IN THE STREAMLINE

Striding in the empty bottom podium, in the space with many thin columns, it is as if you are in a quiet bamboo forest, which is enough for people to relax and relax.



■ 观景流线

行走至空桥结构桥面顶部,可进入两个不同区域和局面的观察平台。从桥的南侧平台可以看到唐河及两侧的农田,

■ **VIEW STREAMLINE**
Walking to the top of the spiral stairs

can enter two viewing platforms of different areas and heights. The viewing platform on the low side can see the sparkling water of the Tang River, and the 1889 Industrial Park on the high side.



■ 驻足流线

这就是国家向下转位，立足于最低端而向上征程，都会使人感觉到不一样的神情。

- [VIEW STREAMLINE](#)

Standing at the top and looking down, stopping at the bottom and looking up at a high place will make people feel a different experience.

■ **入口空间**

该展览厅的剖面为博物馆入口空间。在这里，观众可以清楚地看到博物馆的入口，并体验到博物馆的入口空间。该剖面展示了博物馆的入口空间，并展示了博物馆的入口空间。

■ **Entrance space**

The podium at the bottom is used as entrance space of the overall building where the original cooling tower wall thin columns are preserved. The expo structure and historical materials are used to recall the dusty past and lead them to the inside. New space.

■ 入口空間

这座新址将作为整体建筑的人口空间。在这座深藏腹地的小型居所的墙体和柱列造型、屋檐属的装饰线脚将用历史感的材料带领人们回忆起那个尘封的以往，并带领建筑走向内地的新空间。

- Entrance space

The podium at the bottom is used as the entrance space of the overall building, where the original cooling tower walls and thin columns are preserved. The exposed structure and historical materials lead people to recall the dusty past and lead them to the inside. New space.

● 观察平台

塔身和塔基之间还铺有平整的砖形或块状陶质的观察平台。塔不是垂直和高度可是斜制不同的塔体,阳光从塔身中的缝隙照进塔内,使人在塔身中感到光线透射。

- Viewing platform

between the tower body and the outer skin, the extension of the stair platform forms a viewing platform for leisure and entertainment. Different scenery can be observed from different heights and angles. Sunlight sprinkles mottledly from the gaps in the curtain wall, making people walk through in haste in reality and history.



■ 樓梯空間

利用冷却剂有蒸汽过多装置的需要, 制冰机建造应保从后部板壁推块向上, 即保温水向上的交通阻碍, 也是好建造这世界的最好要素。

- **Stair space**
Utilising the lower

Without too much decoration, the steel spiral staircase winds upwards from the bottom, which is the traffic link for strolling upwards and the key element of relaxing space creation.



● 中庭空間

通过使钢形成的垂直空间还测了冲焊塔的内部结构件。图形的出口高过出神的气态，给人以足够的视觉美。

- Atrium space

The atrium space formed by the spiral staircase restores the original appearance of the cooling tower, and the circular opening creates a sacred atmosphere, giving people enough visual beauty.

唐山

- **唐山风景区**
唐景公园
- **净觉寺**
始建于“平安第一”寺
- **滦州古城**
二千年大周之肇始遗址
- **汉东陵**
秦始皇陵 石炭为形 千代为尊
- **月亮岛**
古称翡翠岛，当风飞絮似雪

CROWD ANALYSIS

[illegible]

建筑表皮

CARBON EMISSION 排碳—固碳
CUTICLE 表皮—衣褶摇曳

主体建筑

CARBON FIXATION
SKIRT SWAYING

观景平台

VIEWING

观景—移步换景
SPIRAL STAIRCASE 旋转楼梯—蜿蜒生长

旋转楼梯

STEP BY STEP TO
CHANGE THE SCENE
WINDING GROWTH

Course Design for Song History Academy at Hebei University

庭影照

宗史小院设计名为“**照影庭**”，取自宋词《生查子·独游雨岩》“**溪边照影行，天在清溪底。天上行行云，人在行云里。**”起伏跌宕的弧面屋顶像是一座座隐秘而又孤高的山峦，结合清晨或傍晚时分从格栅中透出的柔柔暖光，整个小院好似浑然一体，玲珑剔透，静卧在校园的一角。

踩着细碎悠长的木栈道，在雕花窗下吟诵烟雨江南下的诗情画意。而环绕着照影庭的，是纵横交错的水院清溪，漫步于此，悠然惬意。庭内设有书吧沙龙，或两三挚友，或大咖云集，激昂文字，好不惬意！更有回廊折转，讲厅豁然，不禁追忆溥仪先生，创一代帝王传奇，遂作照影庭，映先生风骨，映云上清溪。



生查子·独游雨岩

〔宋〕辛弃疾

溪边照影行，天在清溪底。
天上有行云，人在行云里。
高歌谁和余？空谷清音起。
非鬼亦非仙，一曲桃花水。

照影庭映照，远山之景。
可携琴友二三，或邀宾客同行，共赏庭院之趣。

院落明亮敞然，檐角起伏变化，中有三木成山林，直至夜半，明月高悬，万千夜愿奔涌而来。

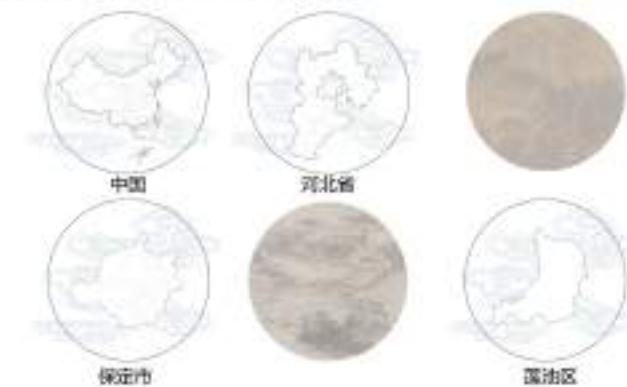
至于白昼时分，孩童嬉笑玩耍，书生挥斥方道，夫子桃李成林，安适祥和之气油然而生。

待酒酣兴尽，看云霁散去，群山复苏，照影庭不见，照影庭犹在。

望云秋水犹在。



地理位置Geographical position



周边环境Surrounding environment



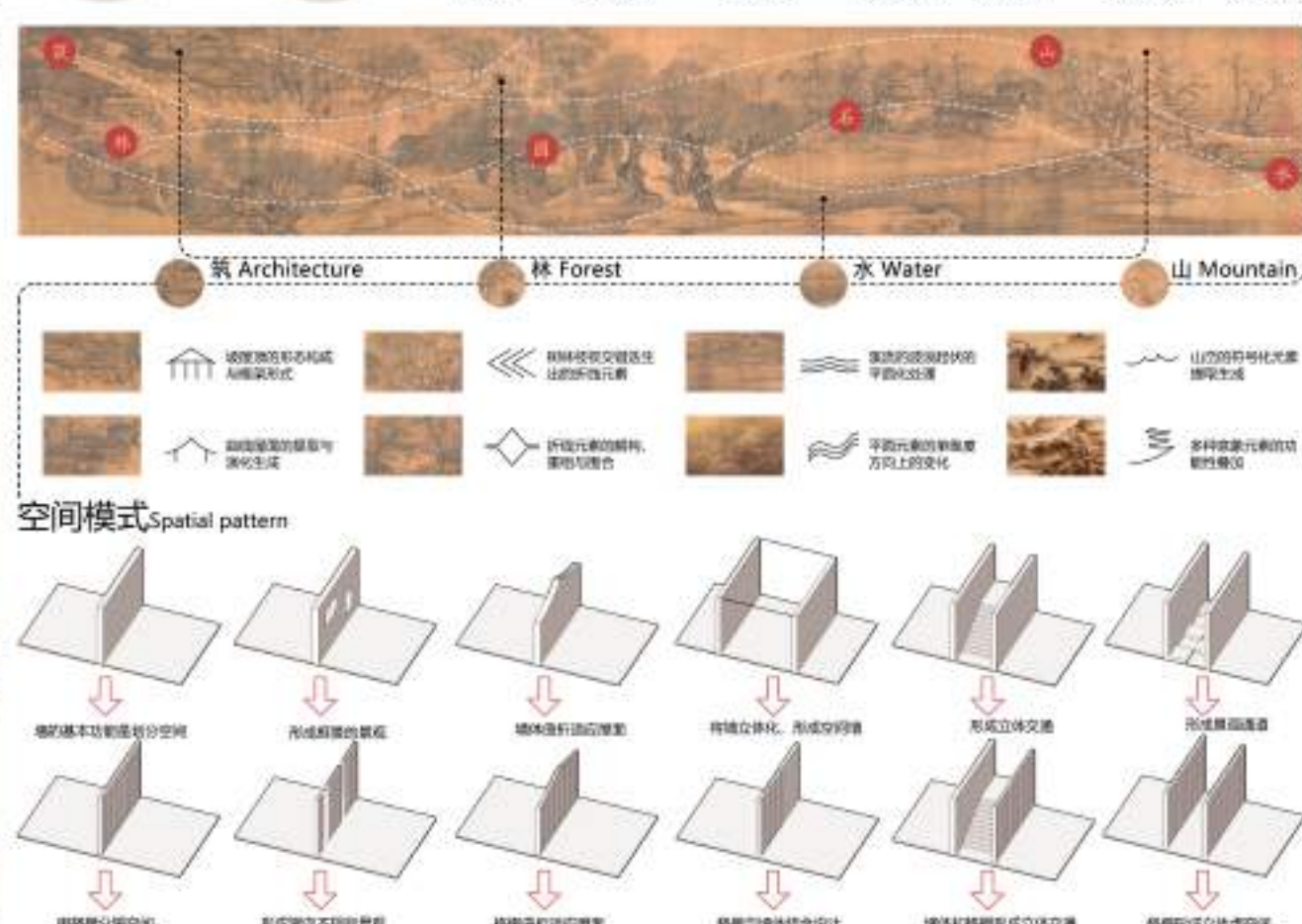
问题和优势 Problems and advantages

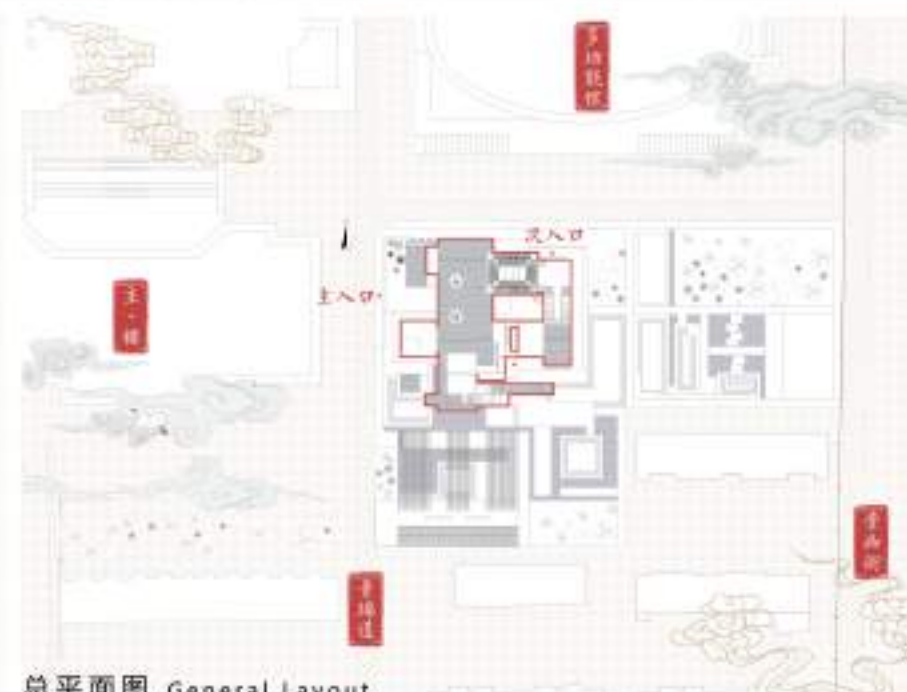
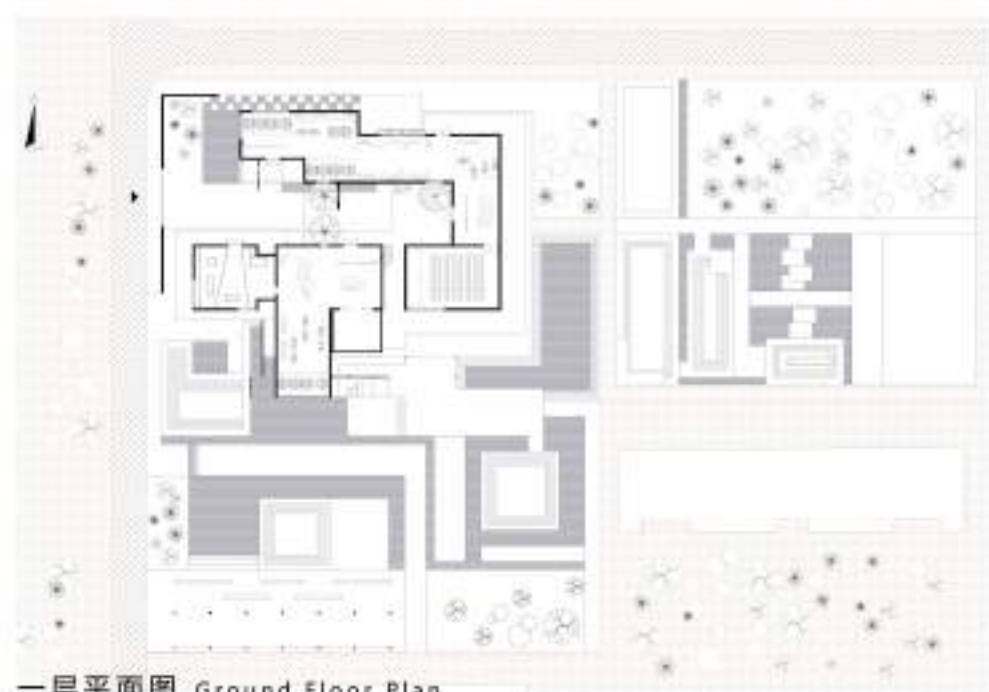
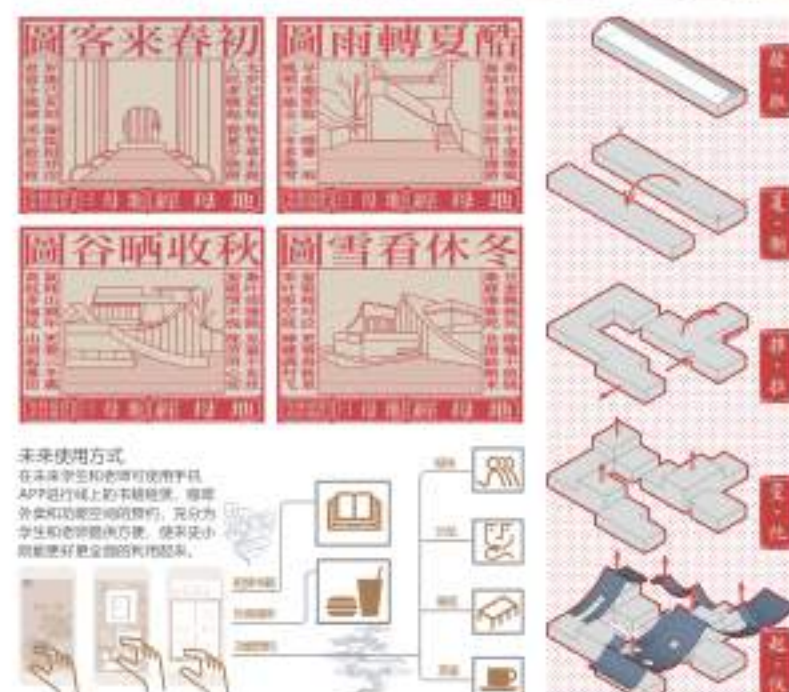
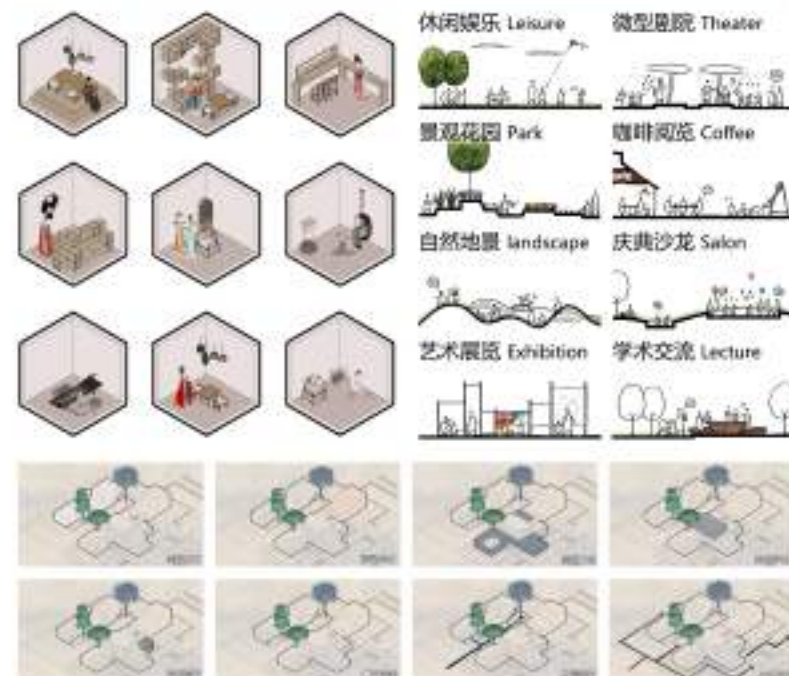
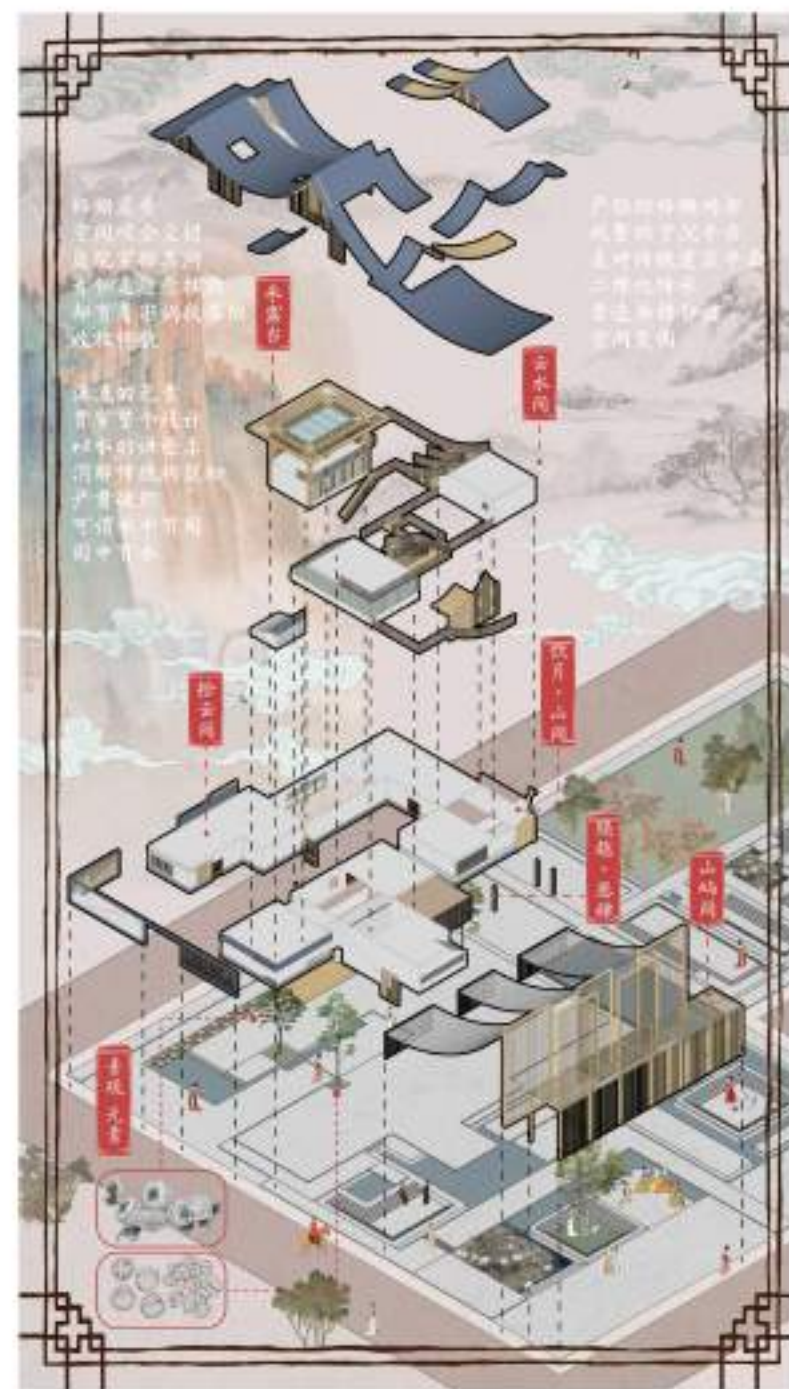


使用人群分析Analysis of users



解决策略Solution strategy



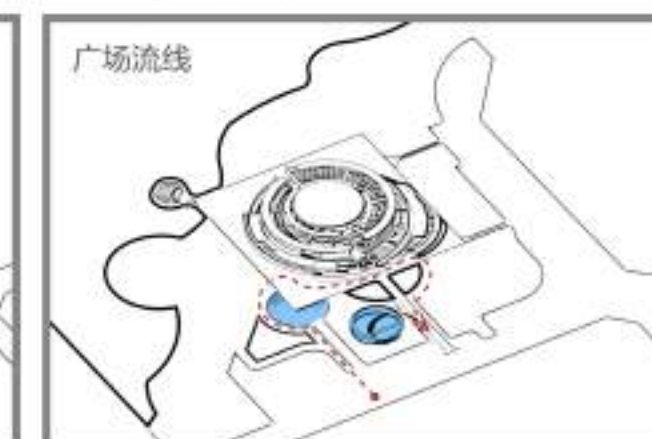
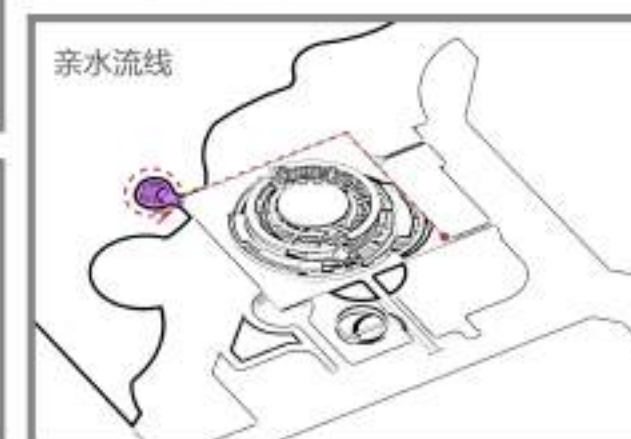
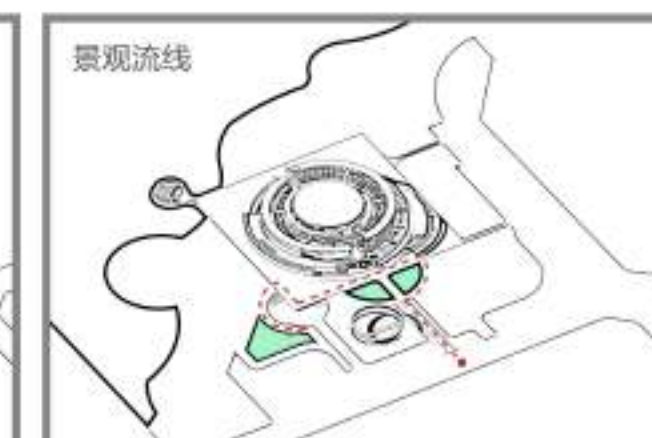
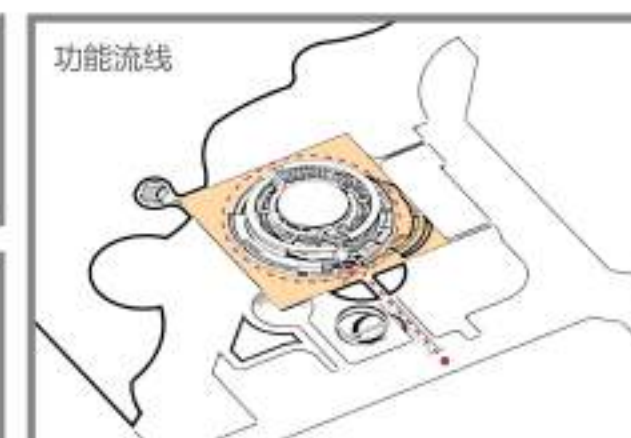
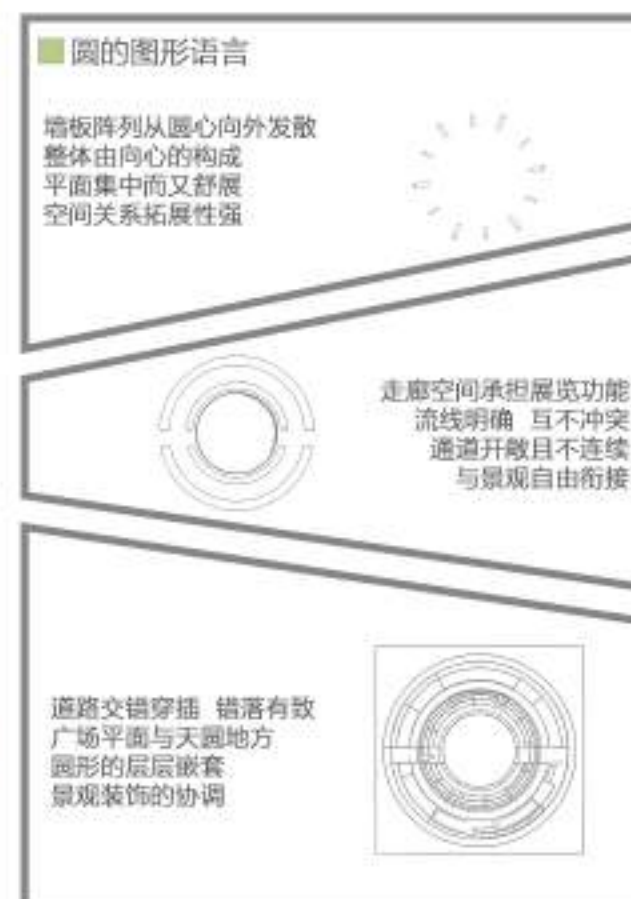
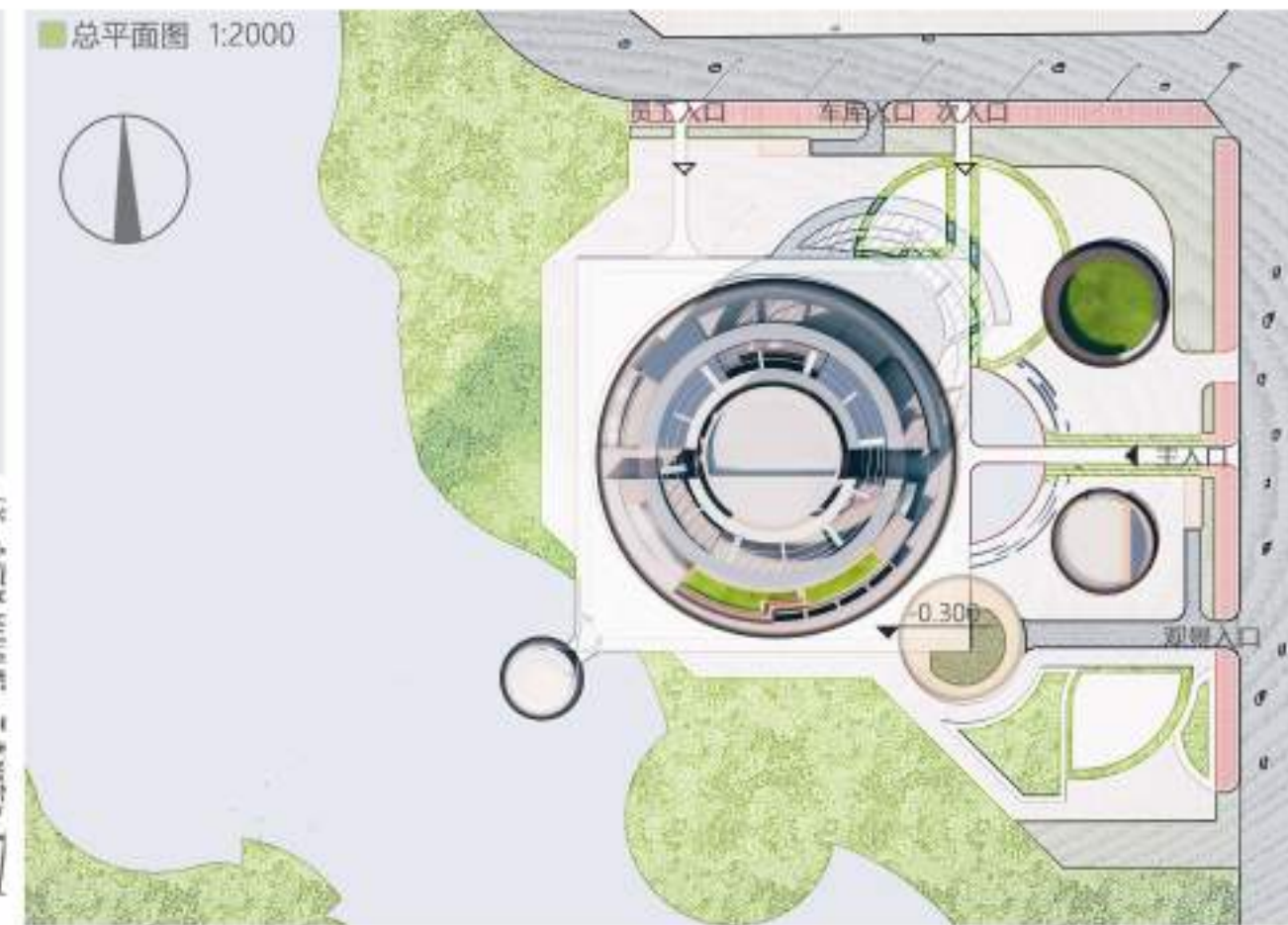


一层平面图 Ground Floor Plan

二层平面图 Floor Plan

总平面图 General Layout

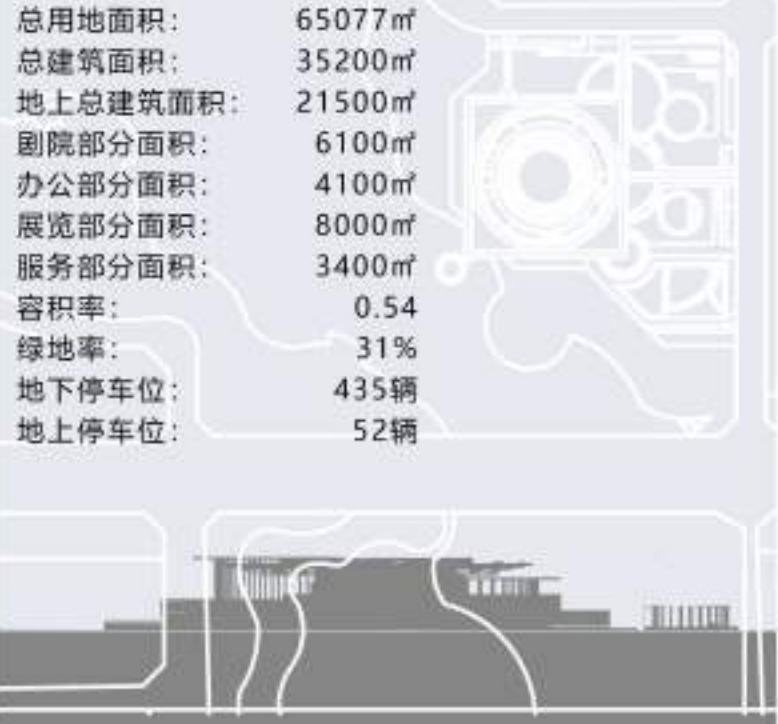
Baoding Convention and Exhibition Center Course Design





■ 经济技术指标

总用地面积:	65077㎡
总建筑面积:	35200㎡
地上总建筑面积:	21500㎡
剧院部分面积:	6100㎡
办公部分面积:	4100㎡
展览部分面积:	8000㎡
服务部分面积:	3400㎡
容积率:	0.54
绿地率:	31%
地下停车位:	435辆
地上停车位:	52辆



十 圓野原

Circulation Nature Theater

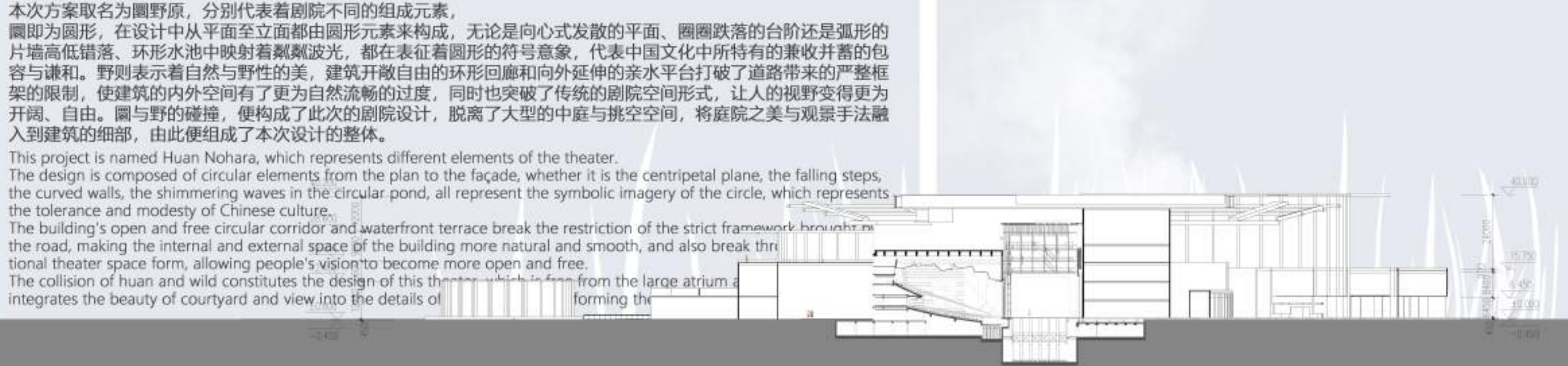


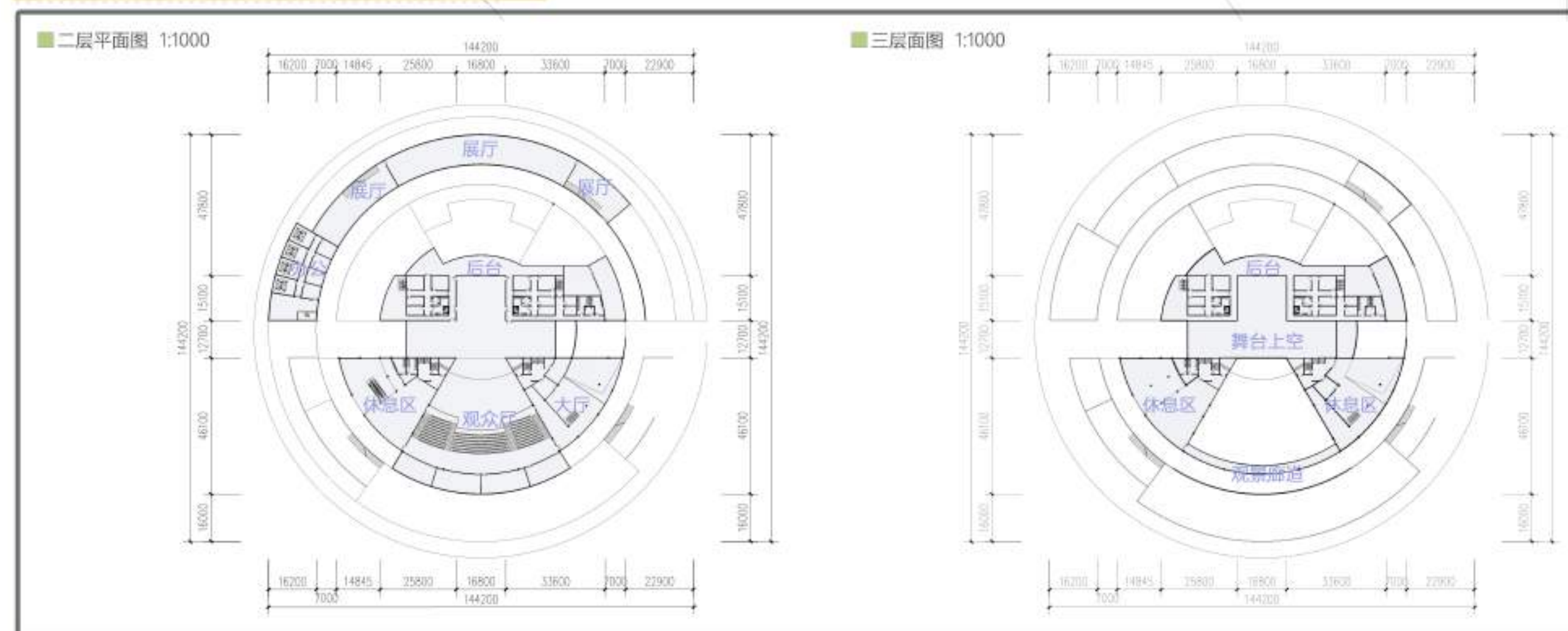
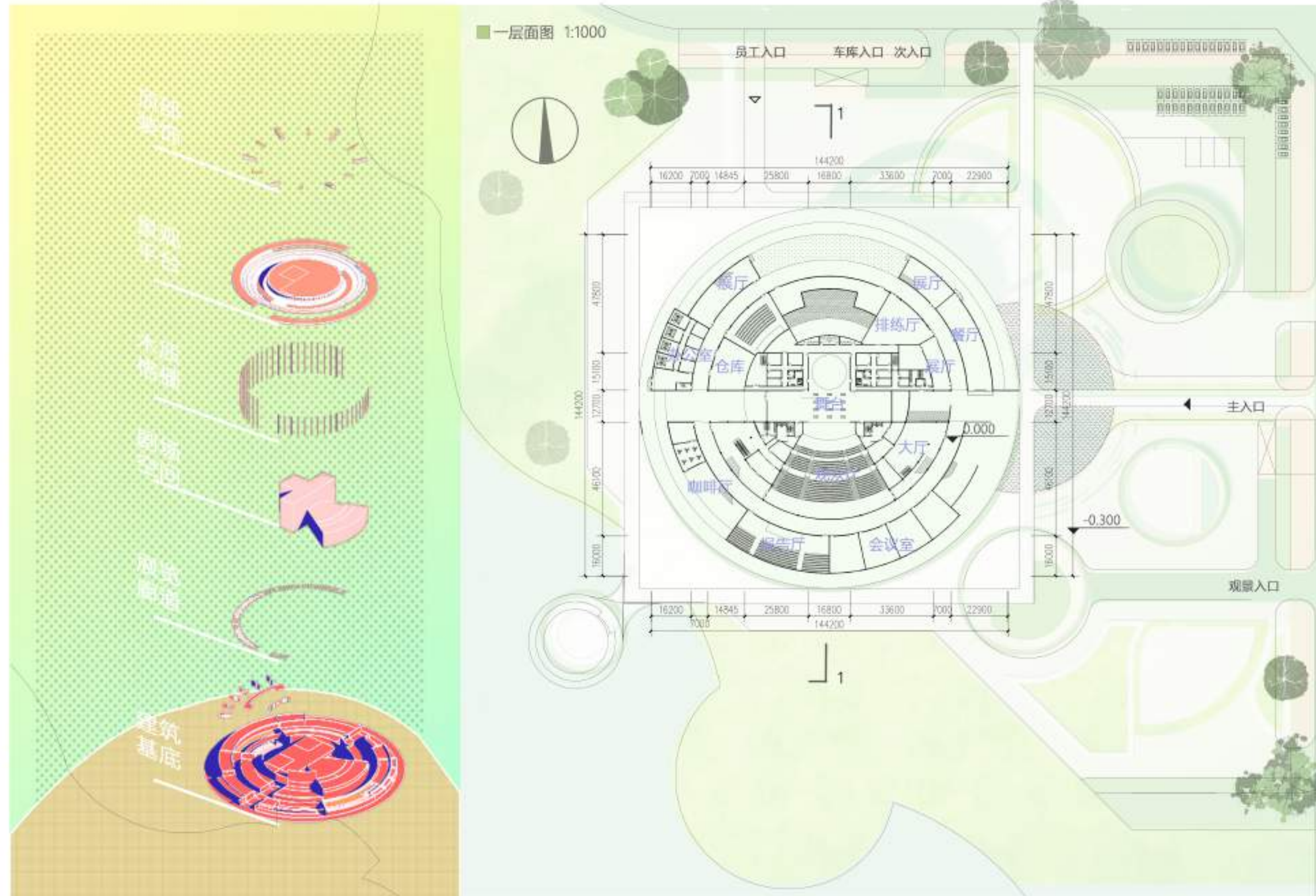
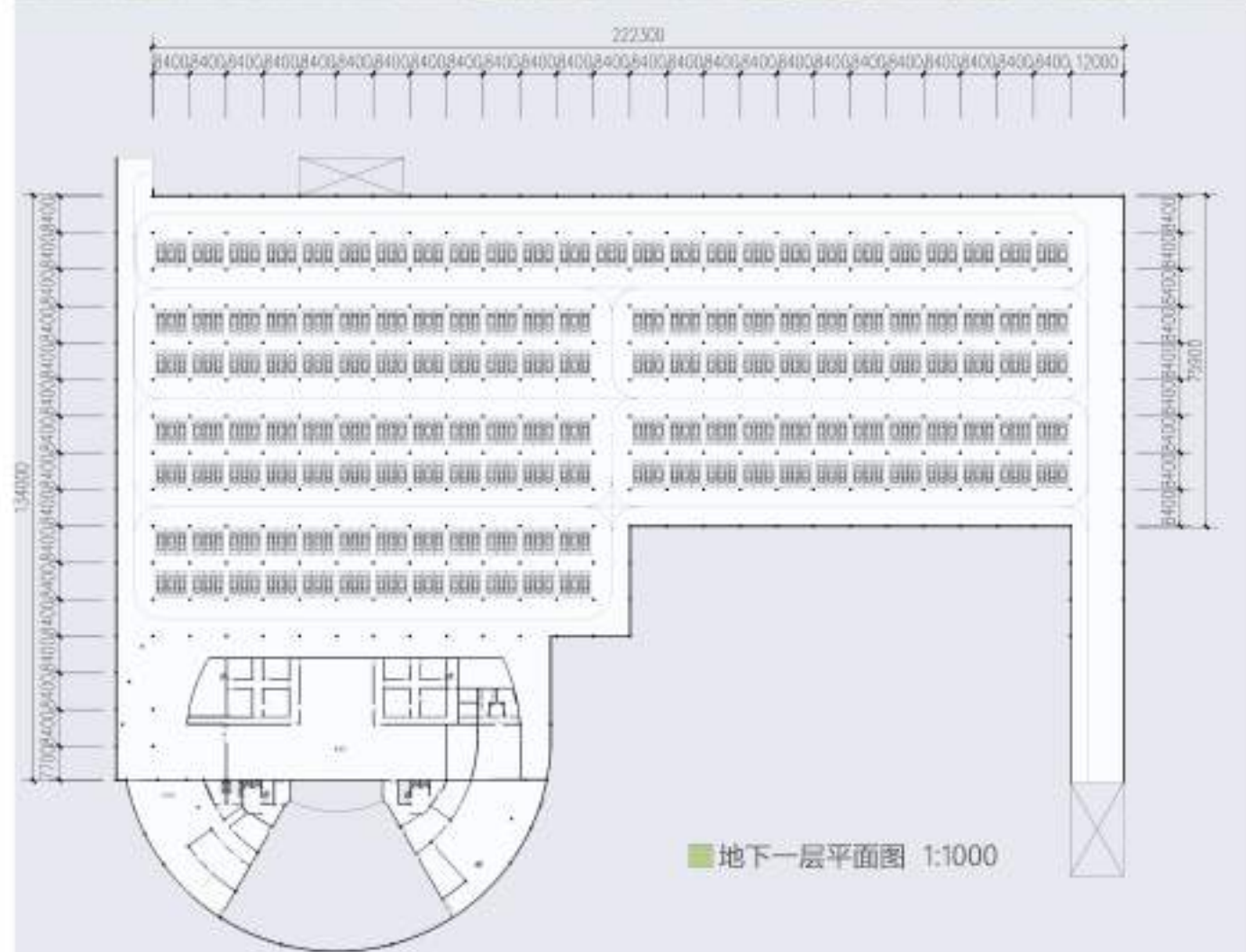
■ 设计说明

本次方案取名为圆野原，分别代表着剧院不同的组成元素，圆即为圆形，在设计中从平面至立面都由圆形元素来构成，无论是向心式发散的平面、圈圈跌落的台阶还是弧形的片墙高低错落、环形水池中映射着粼粼波光，都在表征着圆形的符号意象，代表中国文化中所特有的兼收并蓄的包容与谦和。野则表示着自然与野性的美，建筑开敞自由的环形回廊和向外延伸的亲水平台打破了道路带来的严整框架的限制，使建筑的内外空间有了更为自然流畅的过渡，同时也突破了传统的剧院空间形式，让人的视野变得更为开阔、自由。圆与野的碰撞，便构成了此次的剧院设计，脱离了大型的中庭与挑空空间，将庭院之美与观景手法融入到建筑的细部，由此便组成了本次设计的整体。

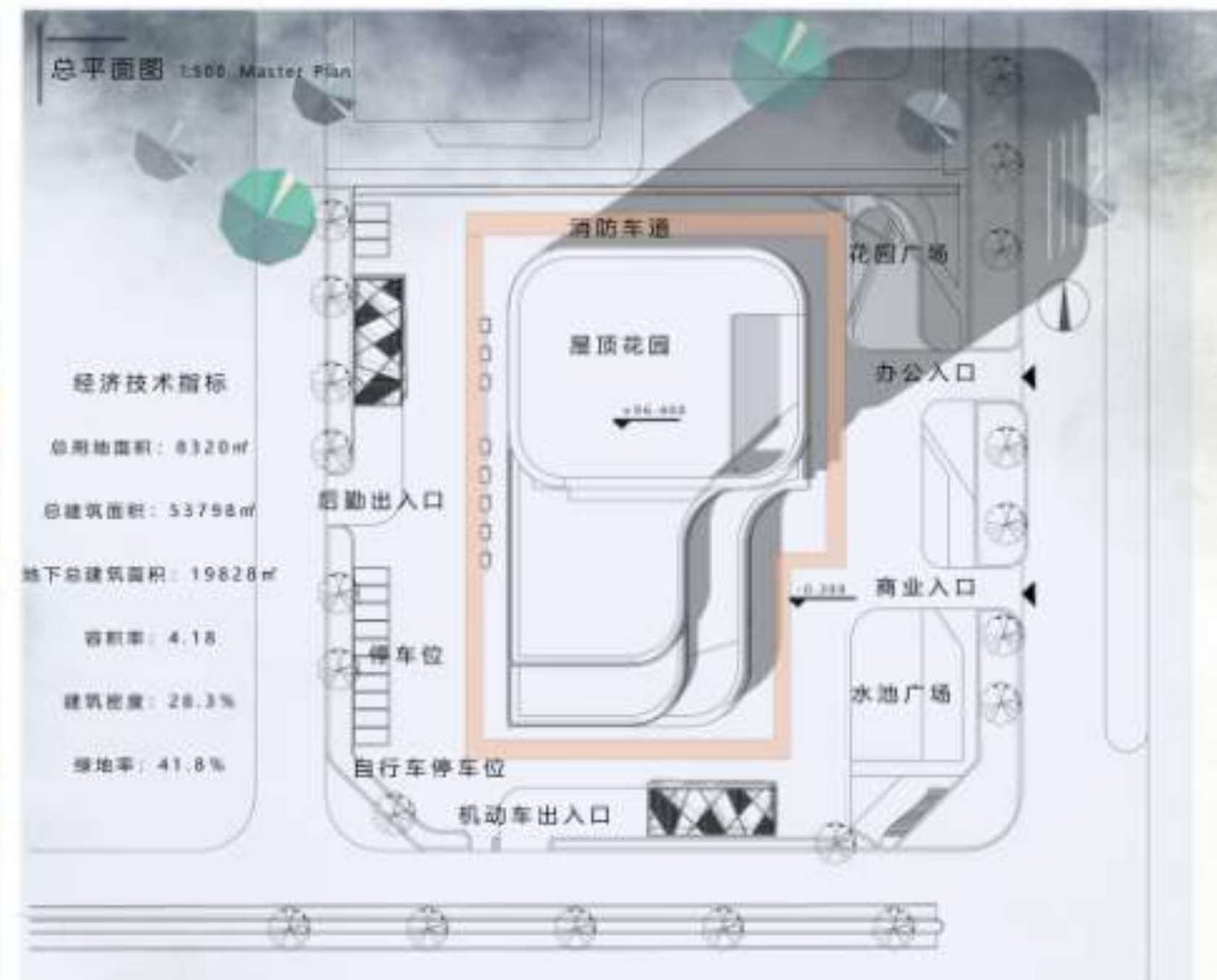
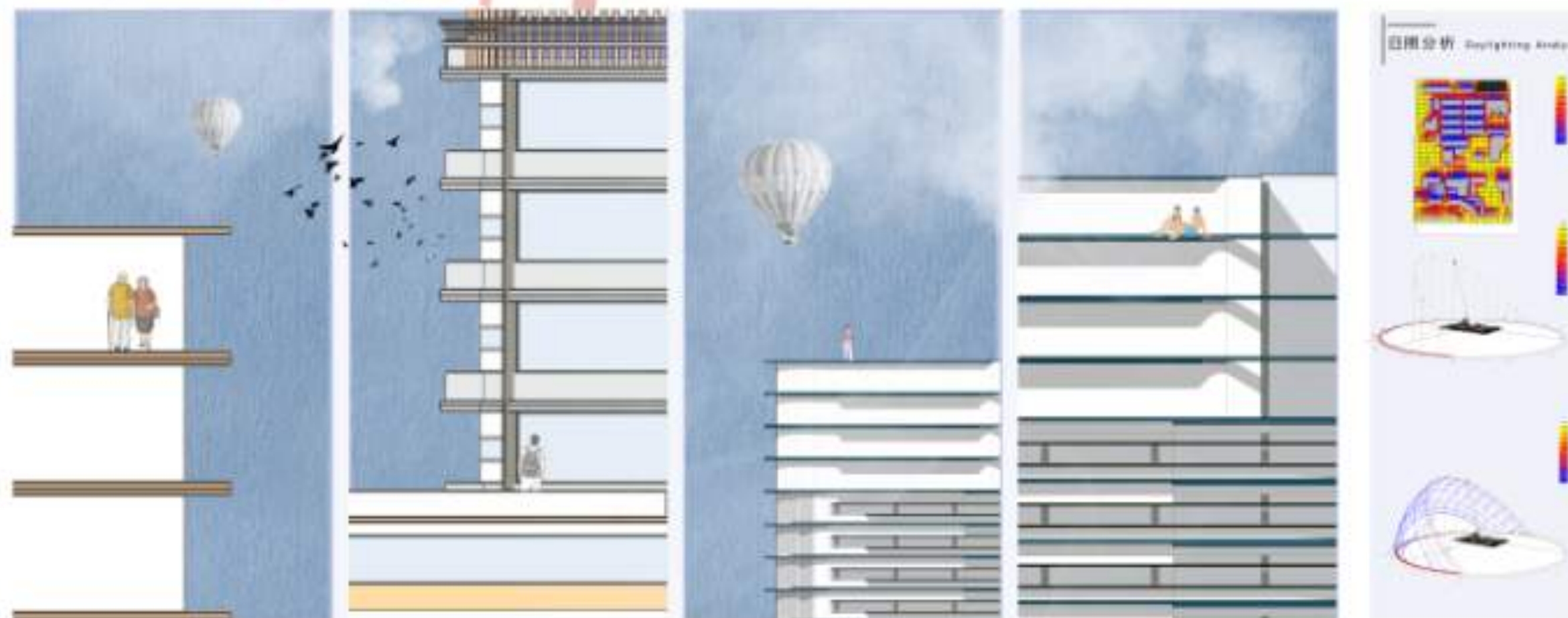
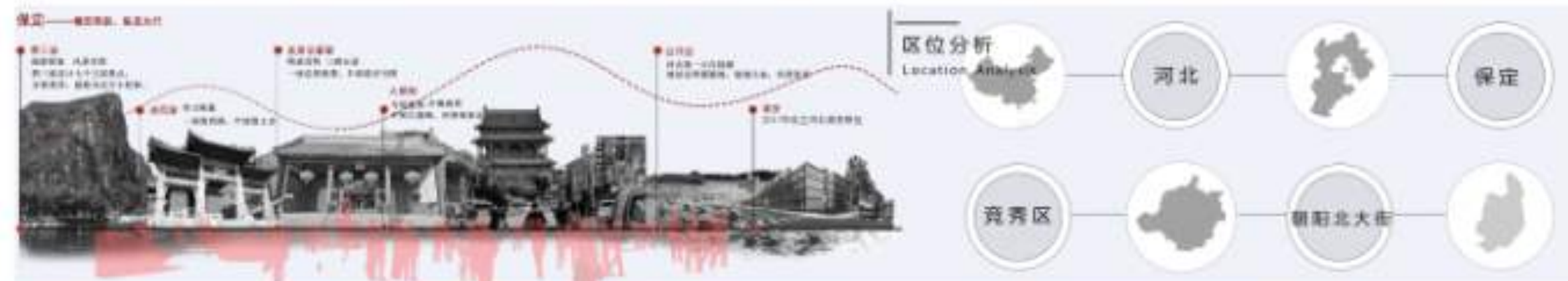
This project is named Huan Nohara, which represents different elements of the theater. The design is composed of circular elements from the plan to the façade, whether it is the centripetal plane, the falling steps, the curved walls, the shimmering waves in the circular pond, all represent the symbolic imagery of the circle, which represents the tolerance and modesty of Chinese culture. The building's open and free circular corridor and waterfront terrace break the restriction of the strict framework brought by the road, making the internal and external space of the building more natural and smooth, and also break traditional theater space form, allowing people's vision to become more open and free. The collision of huan and wild constitutes the design of this theater, which is free from the large atrium and integrates the beauty of courtyard and view into the details of forming the

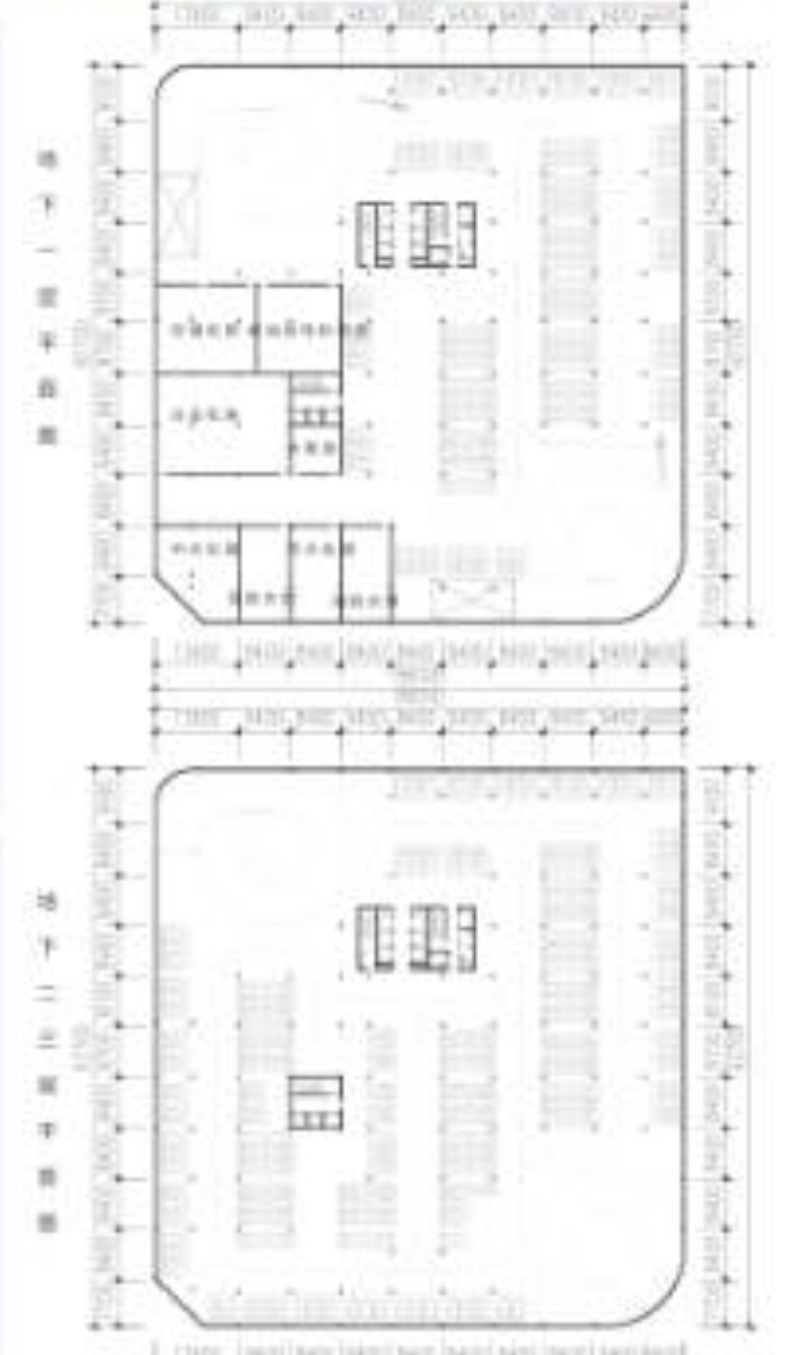
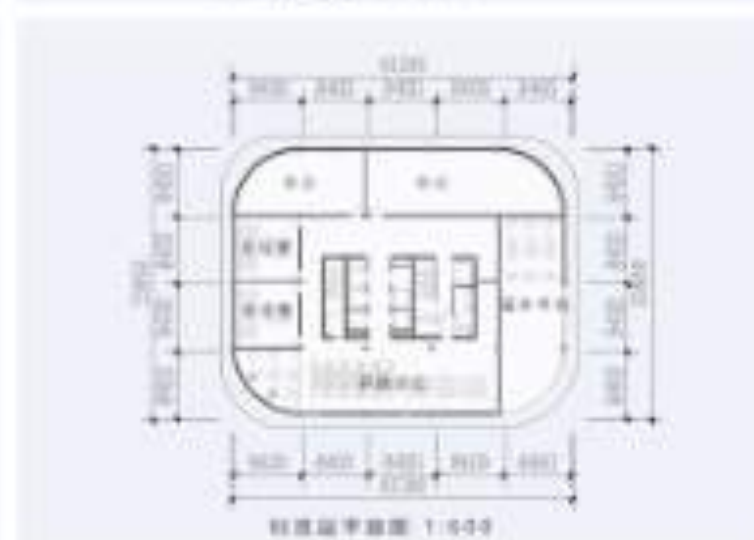
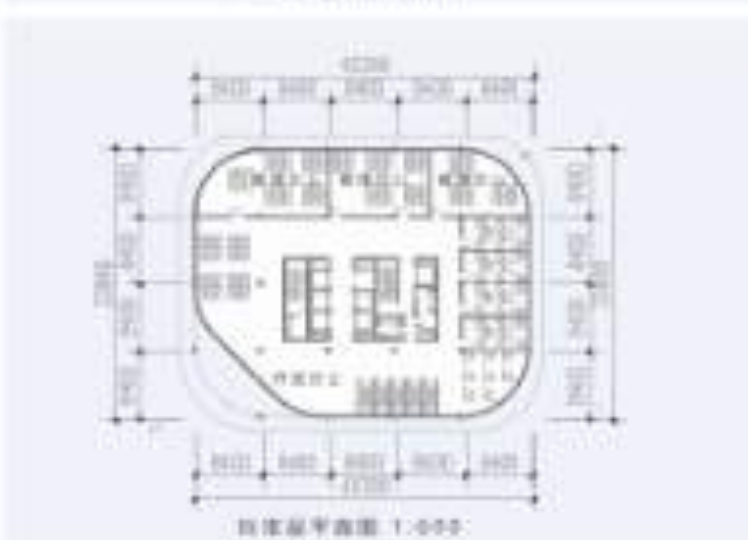
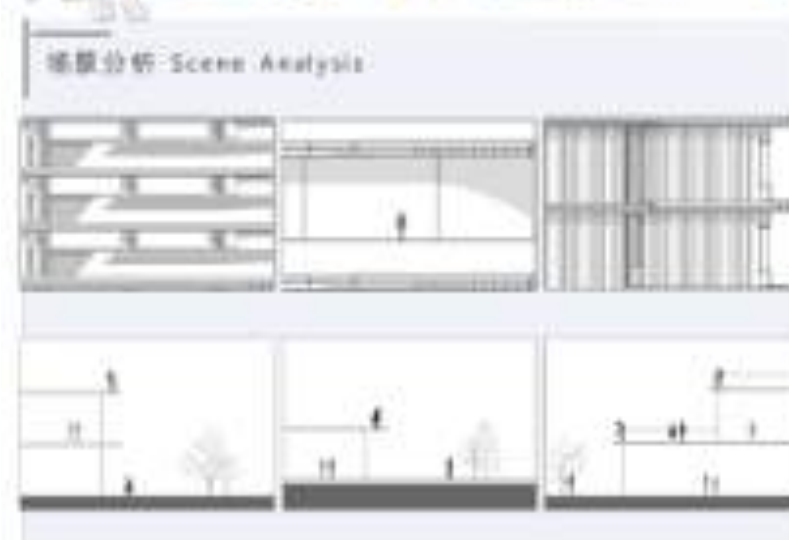
■ 1-1剖面图 1:1000

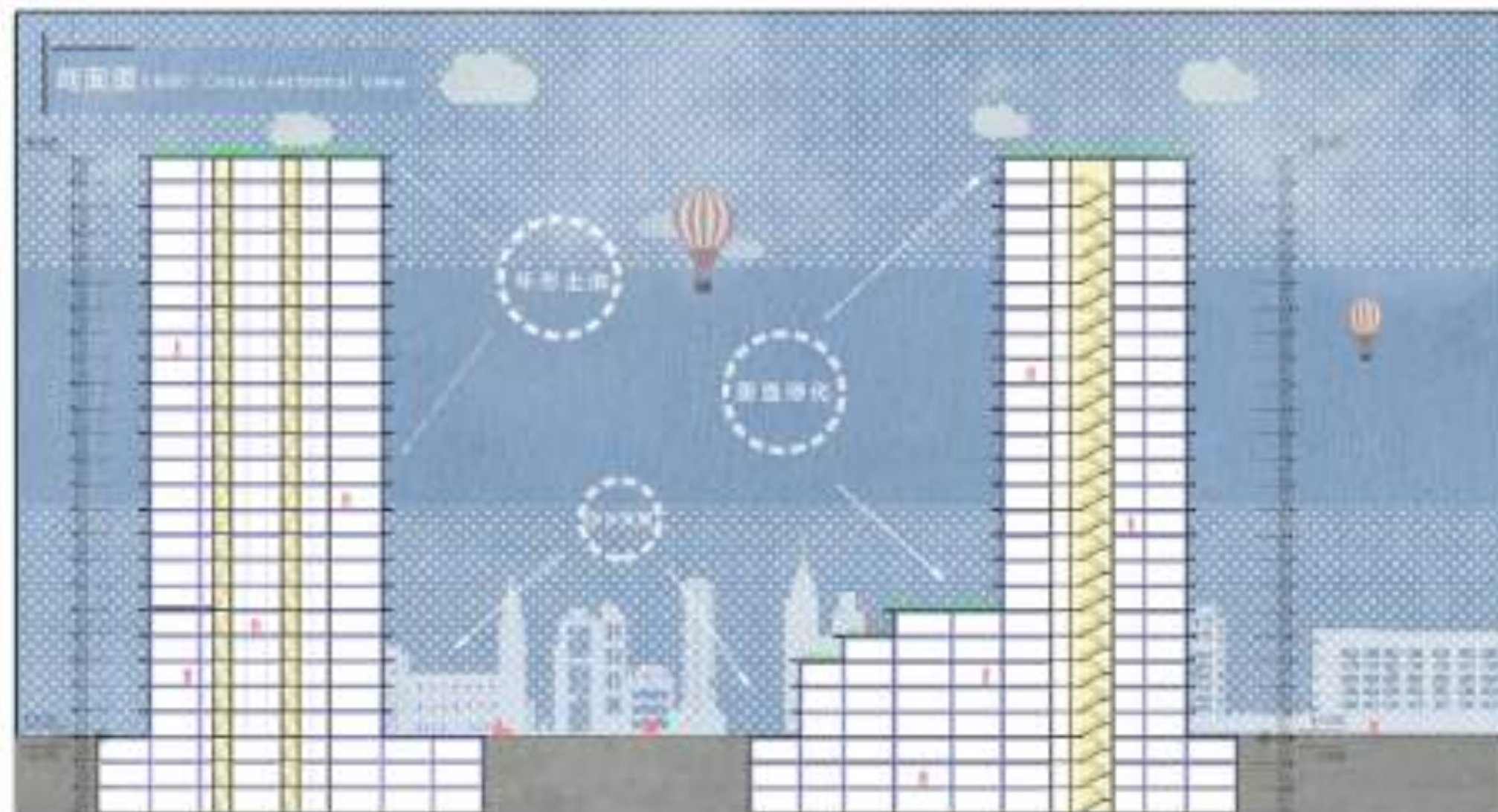
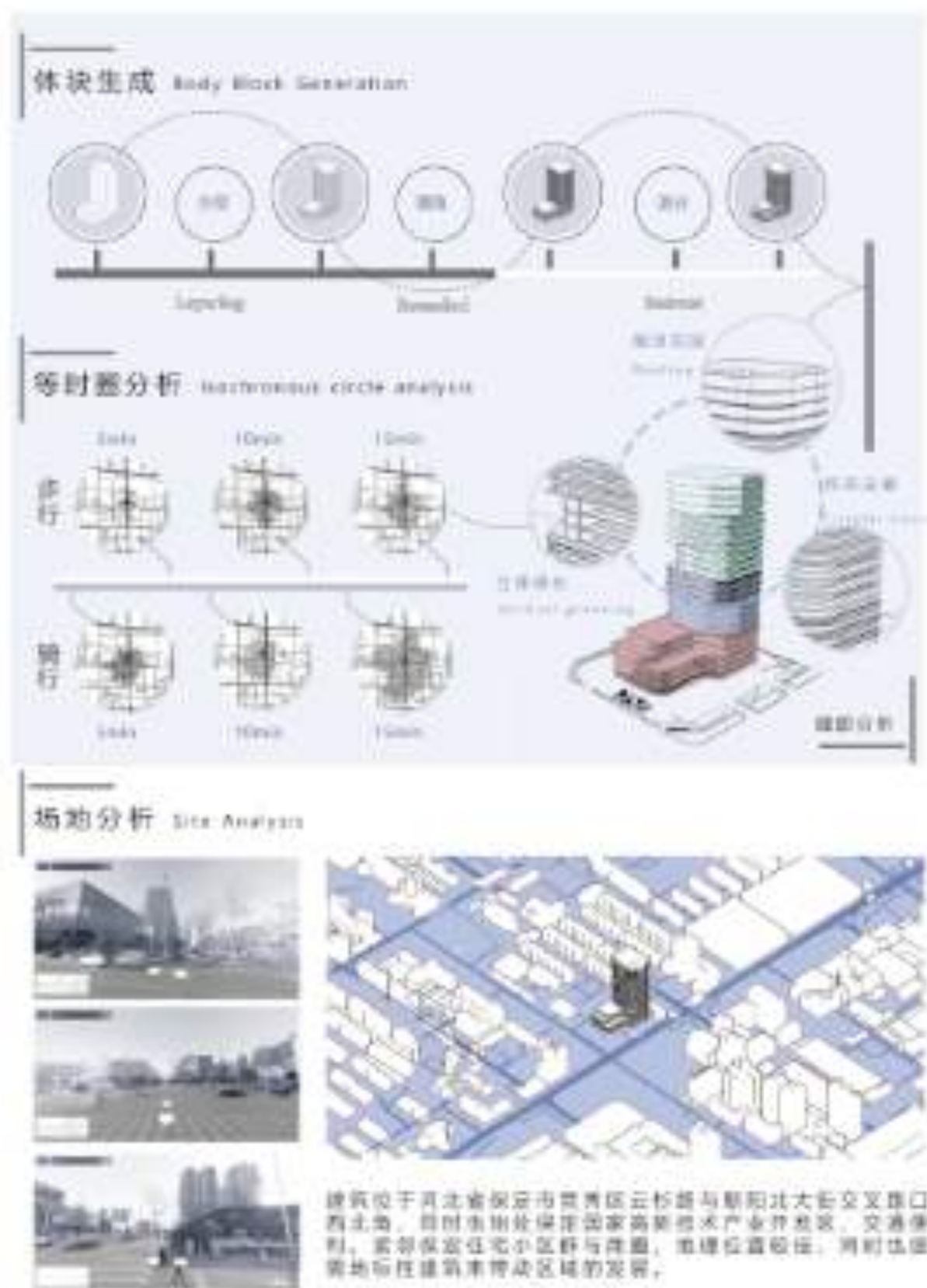




Baoding City High-Rise Office Building Design







设计说明 Design Notes

项目主体建筑位于保定市竞秀区五杉街与朝阳北大街交叉路口西北角，周边有保定国家高新技术产业开发区，交通便利，紧邻保定住宅区与商圈，地理位置优越，同时也带动周边建筑带动区域的发展。

在办公部分建筑高度与周边环境相协调的前提下，为办公提供更为灵活的人员流动与疏散，同时保持建筑上平面的连续性，为建筑带来更多的可能性，同时保持建筑高度与周边环境相协调的前提下，为办公提供更为灵活的人员流动与疏散，同时保持建筑上平面的连续性，为建筑带来更多的可能性。

The main image of the project evolves from the unique geographic location of Baoding, with the Taihang Mountains on the left and the Baiyangdian Waters on the right, combining the different nature and characteristics of the two to create a combination of mountain and water architectural forms, with the mountain peaks to the high-rise office part and the water form in the lower podium part, showing the local nature of Baoding.

In the office part, the stacked and extended circular walkway platform around the building provides more flexible staff flow and vision, while the combination of the planes on specific levels brings a richer and more varied shape to the block, which creates two three-dimensional green platforms in the southwest and southeast directions, lighting up the neighborhood space and bringing more possibilities for the exploration of high-rise landscape buildings in Baoding.

The use of wooden grilles and panels on the facade echoes the traditional identity of Baoding as a historical and cultural city, with history as the background and the future as the direction.



Xiongan New Area Residential Area Design

“我们每个人都渴望属于自己的乌托邦，
在空间的度量与时间的感染下，
找寻内心最终的归宿。”

“我们每个人都渴望属于自己的乌托邦，
在空间的度量与时间的感染下，
找寻内心最终的归宿。”

艺术的源泉。其城市以上海为代表，是典型的中国城市。纽约的露丝·卡普兰(Ruth Freeman)所著的《“美国梦”的破灭》列入首次出版计划之中。在露丝等待这本书有了理想出版商的时候，她突然患了重病并文化大革命下，“白卫兵之地”。



我们采用各种新的形式作为建筑上装饰的构成元素, 这也就是建筑构成的材料语言。把建筑的基本要素, 组合起来, 近而表现出整个建筑的特征或风格, 并进而表现出山水、自然、社会环境、人物、事件又都按照画面构图与造型特性要求, 与建筑当地的文化与风格契合。多元化的因素对居住区内的空间带来了一定的启示。新空间的采用都从不同角度自由地、灵活地多元地表现着, 而门类的建筑知识则从新的多元化形式注入了更多的创造力与想象力, 建筑文化在多元文化背景中得以发展。



单品

- 复制... 集合
- 类型转化
- 展示... 定制



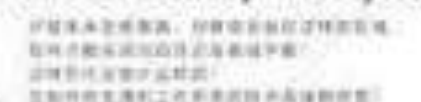
編輯部



在构建中华民族共同体不断发展的历史进程中，铸牢中华民族共同体意识成为不同民族交往、交流、交融的必然要求。铸牢中华民族共同体意识，是维护国家统一、民族团结、社会稳定的必然要求，也是实现中华民族伟大复兴的必然要求。铸牢中华民族共同体意识，是推进国家治理体系和治理能力现代化的必然要求，也是实现中华民族伟大复兴的必然要求。铸牢中华民族共同体意识，是推进国家治理体系和治理能力现代化的必然要求，也是实现中华民族伟大复兴的必然要求。

In a modular building block, the squares explore into different assemblages, which are the latest generation of cellular forms. Within the different assemblages, there is an endless variety of combinations, which are distinguished by the implicit division of walls and the fitting of structures. The final form is on the border of the building, where the space is customized according to the needs of different groups, forming a clear composition that flows from the inside out. The application of assembly technology not only follows the leading edge of construction technology, but also, the technology line implemented by the Ningbo New Area, thus promoting the construction and development of the new area from the perspective of an integrated group and revitalizing the city.

环境体系全面覆盖，经营决策经过特别审核，
 随时调整运营政策以应最坏可能？
 品牌管理是否产品驱动？
 营销推广策略和工具是否随产品品牌调整？



● 物業管理

這本新書以
 簡明扼要的
 方式介紹了
 許多關於
 經濟、管理
 和商業的基
 礎知識。

The book
 presents a
 clear and
 concise
 introduction
 to many
 basic
 concepts
 related to
 economics,
 management,
 and business.



Reconstructing Utopia

重 构 乌 托 邦



Site Planning Analysis

叠层与庭院组合方式为社区提供了富有层次而多样化的视觉感受。环境中心住宅区的两个建筑组团各自承担了不同的功能,但却又彼此紧密联系,创造出和谐而又美好的社区生态环境。

The combination of low-rise and high-rise provides a layered and diverse visual experience for the community. The two groups of buildings surrounding the central residential area each serve different functions, but are closely related to each other, creating a harmoniously balanced and beautiful community ecological environment.

围合的住宅空间形成了较为丰富的社区路网。与高层商住、架空连桥等设施一起构建了具有浓厚生活氛围的社区空间。The enclosed residential spaces form a rich network of streets within the community, and together with ground-floor retail and elevated courtyards, they create a community space with a strong life atmosphere.

立體花圈穿洞+獨立式柱
鐵盤+鐵盤+咬合

閱讀+社區閱讀計畫

doi:10.1017/S0022292412001764

老年住宅区

青年住宅区

中心体室区

图 2-1-10

EDM

立体式的绿化园林空间为社群带来更多时间与空间上的景观与体验。移步换景的行走线路让每一位住户都能够更为全面的亲近自然。

The three-dimensional green garden space brings more time and space to the landscape and experience of the community. The walking routes that change scenery allow each resident to have a more comprehensive affinity with nature.

技术经济指标
TECHNICAL-ECONOMIC INDICATOR

总建筑面积 Total surface area:	257285m ²
建筑用地面积 Building land area:	122559m ²
建筑底面积 Floor area:	44906m ²
容积率 Volume rate:	2.10
覆盖率 Coverage ratio:	0.37
绿地率 Green area ratio:	38.9%
停车位 Parking space:	3000个

對於這套學問，王寧了如指掌。他解釋說，過去中國，都是對皇帝跪，現在對皇帝不跪，對共產黨跪。這不僅是中國政治的進步，也是中國人的進步。為了向中國領導人表示誠意，王寧也開始跪下叩首。

[illegible]

The ground-floor offices catered to the traditional group, and the second-floor, communications and training, was for the new group. The new group included executives, young professionals, performance-based professionals, and a group of people who were doing business and social networking. The building had different entrances for different groups of people.

一些新近發現的證據也支持了這一觀點。例如，加拿大研究人員發現，在一項有關胃病的研究中，那些服用過抑酸藥的人患胃癌的風險比那些沒有服用過的人低。然而，服用抑酸藥的人通常也比那些沒有服用過的人更年輕，這可能導致了誤解。

[illegible]

For example, in the early twentieth century, the economic structure of the agricultural sector in the United States was dominated by large, mechanized farms. These farms were able to produce food more efficiently than smaller, family-owned farms, leading to a decline in the number of small farms. This process was driven by technological innovation and the desire for higher productivity. The result was a concentration of land and resources in the hands of a few large agricultural enterprises.


[网站首页](#)

[illegible]

1. 依序排列，總人數由前至後
 2. 姓名、學號、性別、生日
 3. 性別、生日、學號、姓名
 4. 依序排列，空欄編號，編
 5. 號與性別

1. 請注意姓名、學號、性別、生
 2. 日、姓名及學號、性別、生
 3. 日、學號、生日、姓名、生
 4. 日、姓名、生日、學號、性別

2014年12月15日

[illegible]

网络结构图

通曉之、深明
經濟學、健全
科學能力、事
業成功

公共空间的多重属性

社区环境的开放性

Children

其 通訊、技術、設備、辦公、應

类别：人文类书、素质教育类、少儿类

Comp
Tooth

光電製 科技通訊 交通出行 娛樂

 重要信息
 重要信息

● 中国书画函授大学肇庆分校

Position of the population

Elders

人口老龄化 老年健康服务 现代化农村

Composition of the population

Children Youth Elders

圖書教育 游藝活動 多元文化活動 科技資訊 交通出行 康樂及文化 耆老康樂服務 理財公益

重构乌托邦



The end